

BOUDROUSS RÉDA | CV

AI Engineer | LLM Agents, Verification & Static Analysis

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Availability: from October 2026

Location: Paris, France

Languages: French (Native), English (Professional), Arabic (Conversational)

Profile

At Hexaglobe I industrialized LLM-agent test generation, with sandboxed **multi-agent orchestration**, versioned guardrails, mandatory green tests and human review of every AI output. I also have a **formal-methods** background from Sorbonne University, where I studied **abstract interpretation**, ported the **MOPSA** analyzer (OCaml/C/C++) to WebAssembly and wrote an abstract-interpretation analyzer in Rust. At TrustInSoft I want to combine the two and back AI productivity with sound analysis.

Experience

Quality Analyst Engineer (apprenticeship) - Hexaglobe

2025 – 2026

- Industrialized **AI-assisted development** in an enterprise environment. Built a specification-to-execution loop (PRDs, issues, multi-agent orchestration in Docker sandboxes with planner, implementer and reviewer agents) that produced ~720 integration tests in 3 days, all human-reviewed.
- Wrote the **prompting strategies and guardrails** behind this delegation. Every AI contribution must pass green tests, follow versioned agent rules and stay traceable.
- Wrote ~1,400 **test cases** by hand first (Playwright, Jest, pytest, PHPUnit) for zero-coverage legacy products across 4 stacks (TypeScript, Python, PHP, JS).
- Developed **quality oracles** (VMAF, black-frame/freeze detection, audio analysis) to validate a live video encoder, and shared GitLab CI/CD components (centralized Allure reporting) used by 4+ repositories.

Technical Lead & President (2024–25) - Association Play Sorbonne Université

2023 – 2026

- Designed and operated a virtualized **infrastructure** (Docker servers, CI/CD, networking). As president, led **100+ volunteers** for a cultural event welcoming 10,000+ visitors.

Projects

Reactant analyzer - github.com/rboudrouss/reactant-analyzer

2026

- Wrote a static analyzer in **Rust** (~30k lines, 500+ tests) that catches **React** hook bugs syntactic linters miss, like infinite render loops, derived state and unstable dependencies.
- Evaluates components in abstract domains (intervals, constant sets, reference-stability tags) over a CFG-based IR and reports an infinite render loop when the state fixpoint **widens** instead of converging.
- Resolves and inlines custom hooks across files. The concrete semantics follow the React-tRace paper (OOPSLA 2025).

Mopsa-wasm - github.com/rboudrouss/mopsa-emcc

2026

- Ported **MOPSA**, a sound abstract-interpretation static analyzer for C and Python developed at LIP6 (**OCaml/C/C++**), to **WebAssembly**. Compiled the OCaml runtime and the full native stack (Clang frontend, Apron, GMP, MPFR) into a single static module and replaced the runtime's dynamic FFI loading with a generated table of ~1,400 C primitives.
- Preserved **soundness** in the Wasm environment (fixed rounding mode, no fsetround). Shipped a fully client-side **playground** and a technical **write-up**.

Education

MSc in Computer Science (Software Science & Technology) - Sorbonne University

2024 – 2026

- Coursework: **static analysis by abstract interpretation** and type systems, program specification & **formal verification**, inference of numerical properties, compiler construction, concurrent programming.

BSc in Computer Science - Sorbonne University

2020 – 2024

- Algorithms, computer architecture, operating systems, networks, databases, with team projects in C, Java, OCaml, Python and TypeScript.